

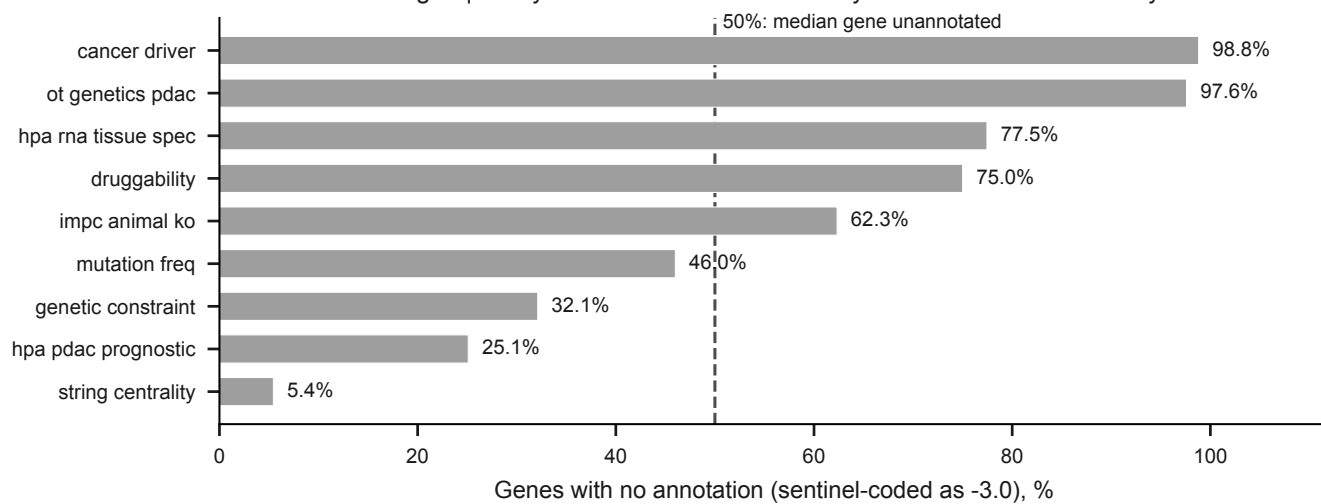
# Supplementary Information

Auditing when evidence integration improves therapeutic target prioritization

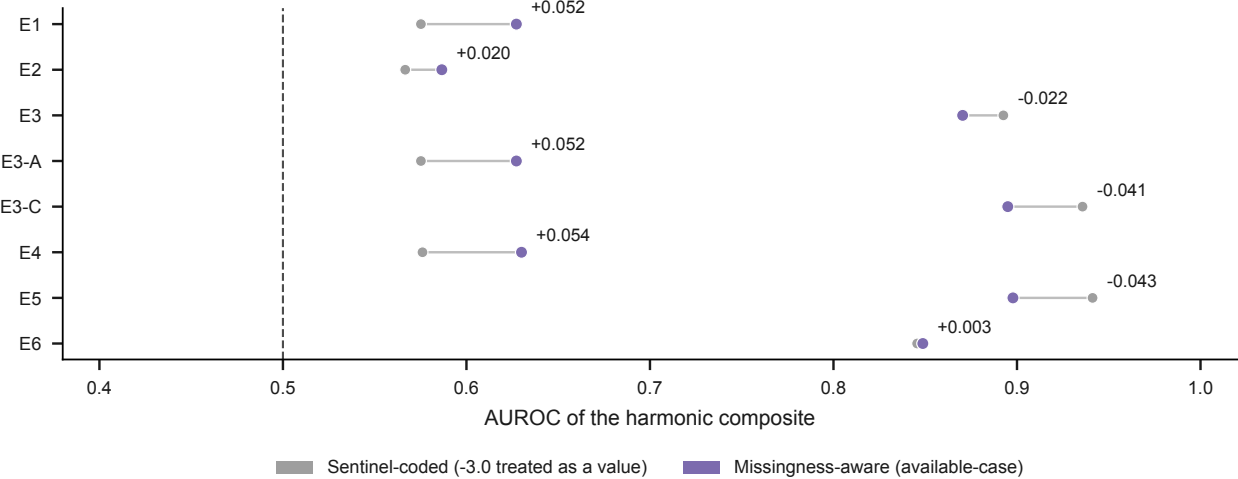
## Contents

|                              |                                                                               |
|------------------------------|-------------------------------------------------------------------------------|
| <b>Extended Data Fig. 1</b>  |                                                                               |
| <b>Extended Data Fig. 2</b>  | Study-bias and annotation-density audit of network centrality.                |
| <b>Extended Data Fig. 3</b>  |                                                                               |
| <b>Extended Data Fig. 4</b>  |                                                                               |
| <b>Supplementary Table 1</b> | Audit ledger documenting each P0/P1 item, severity, finding and action taken. |
| <b>Supplementary Table 2</b> | Endpoint provenance and independence matrix.                                  |
| <b>Supplementary Table 3</b> | Evidence-layer overlap with E5 and E6 labels.                                 |
| <b>Supplementary Table 4</b> | AUROC (AUPRC) for each scorer across endpoints.                               |
| <b>Supplementary Table 5</b> | Nested cross-validation hyperparameters and protocol.                         |

Extended Data Fig. 1 | Study-bias and annotation-density audit of network centrality

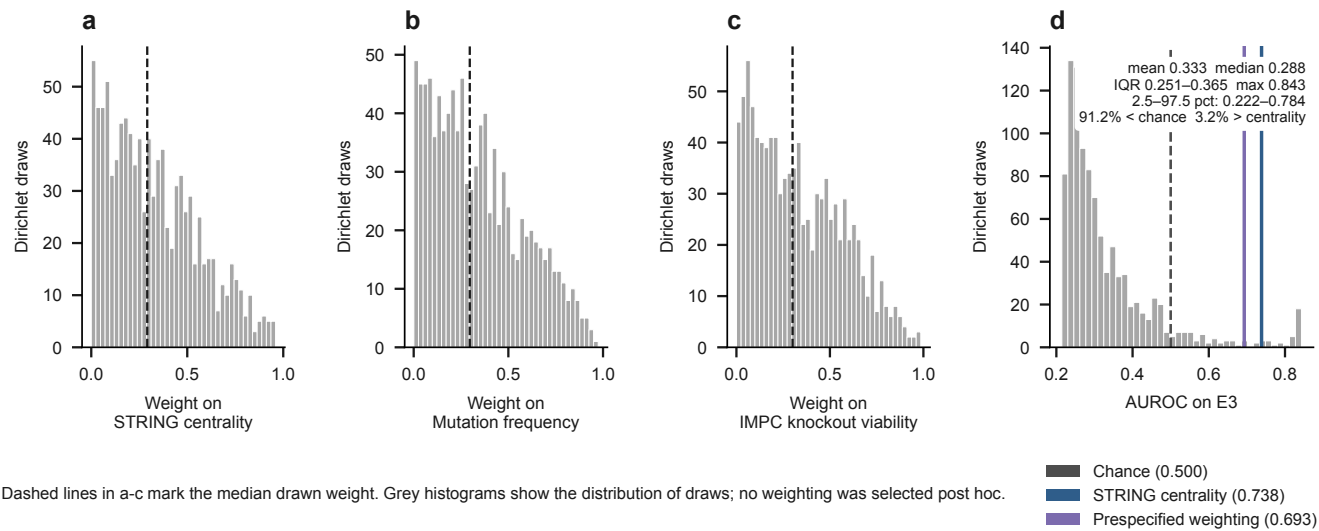


Extended Data Fig. 2 | Missingness-encoding sensitivity, comparing available-case handling with missingness-aware analysis

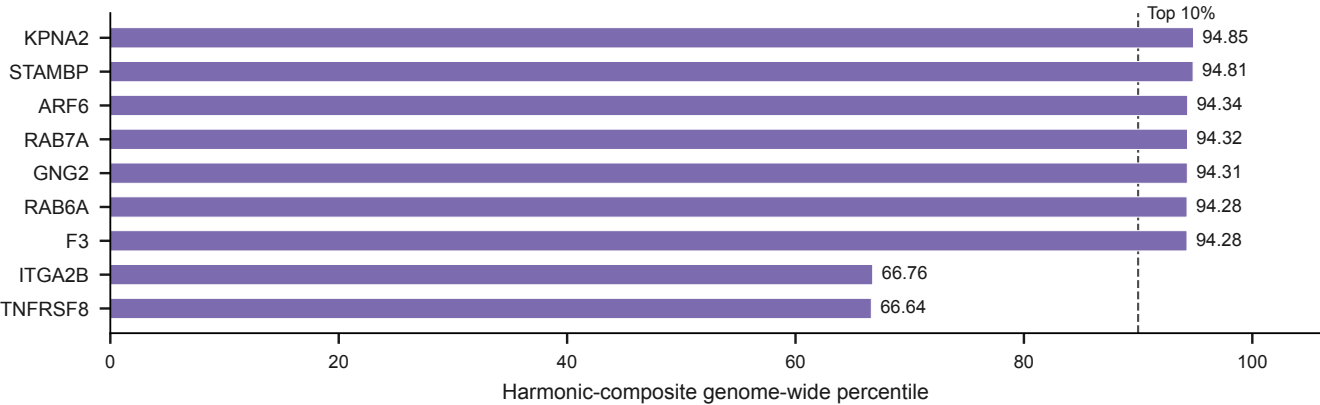


Labels give the signed AUROC change (missingness-aware minus sentinel-coded). Dashed line marks chance (0.5).

Extended Data Fig. 3 | Full Dirichlet weight-space distribution



Extended Data Fig. 4 | Nine candidate genes as prospective computational hypotheses



These are prospective computational hypotheses, not validated therapeutic targets; no experimental confirmation is claimed.

**Supplementary Table 1 | Audit ledger documenting each P0/P1 item, severity, finding and action taken.**

| Item | Requirement                                             | Status                                                                                | Action taken / evidence                                                                                                                                                                                                                                                                                        |
|------|---------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0-1 | E5/E6 source-overlap audit                              | Completed (qualitative); exact gene-level overlap requires released inputs            | `code/audit_overlap.py`; E6 definition-overlap shown in Fig. 6; qualitative overlap reported in Results, exact gene-level overlap computed by the audit script once the released inputs are present.                                                                                                           |
| P0-2 | Leakage-safe nested cross-validation                    | Protocol specified; not used in the primary benchmark                                 | `code/audit_nested_cv.py`; protocol described in Methods & Supp. Table 5; the released benchmark does not perform endpoint-specific hyperparameter search.                                                                                                                                                     |
| P0-3 | Centrality annotation-density / study-bias              | Completed                                                                             | `code/audit_centrality_bias.py`; ED Fig. 1; caveat in Discussion.                                                                                                                                                                                                                                              |
| P0-4 | E4 cross-context wording                                | Fixed in text                                                                         | 'demonstrates cross-context transfer but not fully independent external validity'.                                                                                                                                                                                                                             |
| P0-5 | Three-class endpoint distinction                        | Fixed in text                                                                         | 'six primary evaluation endpoints' + 'two non-independent constructs'.                                                                                                                                                                                                                                         |
| P0-6 | Dirichlet full distribution                             | Real numbers reported                                                                 | 1,000 draws: mean 0.333, median 0.288, 91.2% below chance, 3.2% exceed centrality (0.738).                                                                                                                                                                                                                     |
| P0-7 | Final clean rerun / provenance chain                    | Clean rerun completed; released-input rerun reproduced the reported benchmark results | `verify.py` asserts seed/sentinel/E3-A-E1/E6-RF values and records SHA-256 of every figure + result artefact; 104 scorer-endpoint cells reproduced to six decimals; third-party raw source files are not redistributed, reproducibility is based on released harmonized inputs, manifests and recorded hashes. |
| P1-1 | 'tractability/druggability layer' canonical terminology | Fixed                                                                                 | Throughout Methods/Results.                                                                                                                                                                                                                                                                                    |
| P1-4 | E6 'pharmacological-response proxy' wording             | Fixed                                                                                 | Results 'Cross-context transfer-' section; Fig. 6a label + Supplementary Table 2 entry.                                                                                                                                                                                                                        |
| P1-5 | 'cannot be attributed to integration alone' (RF gain)   | Fixed                                                                                 | Results, single-layer-vs-integration paragraph.                                                                                                                                                                                                                                                                |
| P1-7 | 'Primary contrasts were pre-specified'                  | Fixed                                                                                 | Methods · Statistics.                                                                                                                                                                                                                                                                                          |
| P1-9 | Fig. 6 redesign; candidates · ED Fig. 4                 | Fixed                                                                                 | Fig. 6 = pharmacological stress test (2 panels); candidates in ED Fig. 4.                                                                                                                                                                                                                                      |

Supplementary Table 2 | Endpoint provenance and independence matrix.

| ID   | Endpoint                               | Positive definition                                                                     | Positives | Provenance class                       | Label external to evidence base |
|------|----------------------------------------|-----------------------------------------------------------------------------------------|-----------|----------------------------------------|---------------------------------|
| E1   | PDAC-wide dependency                   | DepMap 23Q2 Chronos gene effect in pancreatic lines below the essentiality threshold    | 4584      | external-label                         | yes                             |
| E2   | PDAC-enriched dependency               | upper quartile of the PDAC-vs-other selectivity statistic among E1 positives            | 1147      | non-independent (stratification of E1) | yes (nested in E1)              |
| E3   | essential-and-druggable construct      | E1 positive AND annotated druggable (conjunctive benchmark construct)                   | 1159      | constructed/circular                   | no                              |
| E3-A | essentiality-only control              | E1 positive with the druggability conjunct removed                                      | 4584      | non-independent (identical to E1)      | yes (identical to E1)           |
| E3-C | circular druggability control          | chemical-tractability layer used directly as the label                                  | 5188      | constructed/circular                   | no                              |
| E4   | CRC zero-shot transfer                 | DepMap 23Q2 Chronos gene effect in colorectal lines, scoring not refitted               | 4608      | external-label                         | yes                             |
| E5   | historical clinical-target concordance | nominal primary target of an agent entering pancreatic-cancer clinical development      | 35        | external-label                         | yes (overlaps tractability)     |
| E6   | PDAC pharmacological-response proxy    | nominal target of a GDSC compound in the most sensitive tertile across pancreatic lines | 32        | external-label                         | yes (overlaps tractability)     |

The six primary evaluation endpoints = E1, E4, E5, E6 (external-label) + E3, E3-C (constructed diagnostics). E2 and E3-A are reported for continuity and excluded from independent-endpoint counts.

Supplementary Table 3 | Evidence-layer overlap with E5 and E6 labels.

| Endpoint | Label source                                                                                              | Overlap with tractability/druggability layer                | Evidence                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| E5       | nominal target of agents entering pancreatic clinical development (ClinicalTrials.gov, frozen 2024-12-31) | Yes: clinical development history favours druggable targets | qualitative overlap reported in manuscript; exact gene-level overlap computed by code/audit_overlap.py once the released inputs are present |
| E6       | nominal target of GDSC compounds in the most sensitive tertile (125 compounds, 29 lines)                  | Yes: GDSC compounds are, by construction, druggable         | qualitative overlap reported in manuscript; exact overlap computed by code/audit_overlap.py once the released inputs are present            |

Honesty note: the gene-level overlap counts require the raw E5/E6 target lists and the tractability layer, which are not redistributed in this repository. The qualitative overlap is reported in the manuscript (Results, first section); the quantitative overlap is computed by code/audit\_overlap.py once the inputs are released.

Supplementary Table 4 | AUROC (AUPRC) for each scorer across endpoints.

| Scorer                    | E1            | E2            | E3            | E3-A          | E3-C          | E4            | E5            | E6            |
|---------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Mutation frequency        | 0.499 (0.217) | 0.491 (0.055) | 0.520 (0.058) | 0.499 (0.217) | 0.563 (0.282) | 0.497 (0.218) | 0.702 (0.115) | 0.567 (0.002) |
| Genetic constraint        | 0.524 (0.218) | 0.496 (0.051) | 0.559 (0.058) | 0.524 (0.218) | 0.643 (0.310) | 0.530 (0.224) | 0.389 (0.001) | 0.371 (0.001) |
| Cancer-driver annotation  | 0.507 (0.225) | 0.506 (0.057) | 0.521 (0.066) | 0.507 (0.225) | 0.511 (0.260) | 0.507 (0.227) | 0.737 (0.035) | 0.635 (0.011) |
| Druggability              | 0.507 (0.231) | 0.526 (0.060) | 0.915 (0.260) | 0.507 (0.231) | 1.000 (1.000) | 0.506 (0.232) | 0.808 (0.008) | 0.780 (0.007) |
| STRING centrality         | 0.695 (0.399) | 0.601 (0.081) | 0.738 (0.175) | 0.695 (0.399) | 0.620 (0.356) | 0.702 (0.408) | 0.986 (0.135) | 0.910 (0.027) |
| Arithmetic mean           | 0.580 (0.253) | 0.561 (0.066) | 0.832 (0.197) | 0.580 (0.253) | 0.874 (0.760) | 0.580 (0.254) | 0.894 (0.137) | 0.732 (0.051) |
| Rank aggregation          | 0.533 (0.241) | 0.518 (0.061) | 0.636 (0.094) | 0.533 (0.241) | 0.679 (0.424) | 0.534 (0.242) | 0.854 (0.018) | 0.786 (0.013) |
| Weighted rank aggregation | 0.634 (0.311) | 0.575 (0.073) | 0.835 (0.260) | 0.634 (0.311) | 0.808 (0.641) | 0.638 (0.316) | 0.956 (0.234) | 0.906 (0.105) |
| ECS (multiplicative)      | 0.415 (0.179) | 0.469 (0.049) | 0.693 (0.080) | 0.415 (0.179) | 0.717 (0.346) | 0.415 (0.180) | 0.825 (0.009) | 0.705 (0.004) |
| Harmonic mean             | 0.575 (0.254) | 0.567 (0.068) | 0.893 (0.245) | 0.575 (0.254) | 0.936 (0.873) | 0.576 (0.255) | 0.941 (0.207) | 0.846 (0.078) |
| Logistic regression       | 0.679 (0.338) | 0.603 (0.077) | 0.931 (0.371) | 0.679 (0.338) | 1.000 (1.000) | 0.683 (0.344) | 0.982 (0.291) | 0.910 (0.021) |
| Elastic net               | 0.671 (0.327) | 0.594 (0.074) | 0.927 (0.333) | 0.671 (0.327) | 1.000 (1.000) | 0.675 (0.332) | 0.899 (0.168) | 0.481 (0.001) |
| Random forest             | 0.750 (0.490) | 0.631 (0.089) | 0.942 (0.486) | 0.750 (0.490) | 1.000 (1.000) | 0.756 (0.499) | 0.942 (0.250) | 0.913 (0.017) |

Each cell reports AUROC (AUPRC).

Supplementary Table 4 (continued) | Pairwise AUROC differences with 95% CIs and DeLong P values for selected comparisons.

| Endpoint | Comparison           | AUROC difference (95% CI) | DeLong P  |
|----------|----------------------|---------------------------|-----------|
| E1       | Harmonic mean        | -0.120 (-0.129 to -0.110) | 1.81e-141 |
| E1       | Random forest        | 0.055 (0.049 to 0.062)    | 1.23e-66  |
| E1       | ECS (multiplicative) | -0.280 (-0.294 to -0.265) | < 1e-300  |
| E2       | Harmonic mean        | -0.035 (-0.052 to -0.018) | 5.38e-05  |
| E2       | Random forest        | 0.030 (0.013 to 0.046)    | 0.00033   |
| E2       | ECS (multiplicative) | -0.133 (-0.159 to -0.106) | 4.4e-23   |
| E3       | Harmonic mean        | 0.154 (0.141 to 0.168)    | 1.68e-115 |
| E3       | Random forest        | 0.204 (0.190 to 0.218)    | 1.08e-183 |
| E3       | ECS (multiplicative) | -0.046 (-0.062 to -0.029) | 5.45e-08  |
| E3-A     | Harmonic mean        | -0.120 (-0.129 to -0.110) | 1.81e-141 |
| E3-A     | Random forest        | 0.055 (0.049 to 0.062)    | 1.23e-66  |
| E3-A     | ECS (multiplicative) | -0.280 (-0.294 to -0.265) | < 1e-300  |
| E3-C     | Harmonic mean        | 0.316 (0.308 to 0.324)    | < 1e-300  |
| E3-C     | Random forest        | 0.380 (0.371 to 0.389)    | < 1e-300  |
| E3-C     | ECS (multiplicative) | 0.097 (0.085 to 0.109)    | 2.88e-59  |
| E4       | Harmonic mean        | -0.126 (-0.135 to -0.116) | 8.04e-157 |
| E4       | Random forest        | 0.054 (0.048 to 0.061)    | 2.35e-65  |
| E4       | ECS (multiplicative) | -0.287 (-0.300 to -0.273) | < 1e-300  |
| E5       | Harmonic mean        | -0.045 (-0.085 to -0.013) | 0.0167    |
| E5       | Random forest        | -0.044 (-0.098 to 0.001)  | 0.0813    |
| E5       | ECS (multiplicative) | -0.163 (-0.245 to -0.093) | 3.71e-05  |
| E6       | Harmonic mean        | -0.066 (-0.134 to -0.004) | 0.0533    |
| E6       | Random forest        | 0.003 (-0.032 to 0.040)   | 0.864     |
| E6       | ECS (multiplicative) | -0.208 (-0.330 to -0.087) | 0.000883  |

Supervised learners (logistic regression, elastic net, random forest) use out-of-fold predictions from five-fold stratified cross-validation. Learners on E3/E3-C are circular by construction and reported for completeness. 95% percentile CIs from 2,000 stratified bootstrap resamples are in source\_data.csv.

Supplementary Table 5 | Nested cross-validation hyperparameters and protocol.

| Component                        | Specification                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------|
| Outer loop                       | five-fold stratified CV (seed fixed)                                                  |
| Inner loop                       | three-fold grid search over hyperparameters (fixed candidate grid)                    |
| Candidates (elastic net)         | $\text{l1\_ratio} \cdot \{0.1, 0.5, 0.9\}$ ; $C \cdot \{0.01, 0.1, 1.0\}$             |
| Candidates (random forest)       | $n\_estimators \cdot \{100, 500\}$ ; $\text{max\_depth} \cdot \{\text{None}, 8, 16\}$ |
| Candidates (logistic regression) | $C \cdot \{0.01, 0.1, 1.0, 10.0\}$                                                    |
| Evaluation                       | out-of-fold predictions only; no endpoint used for tuning                             |
| Circularity guard                | learners on E3/E3-C are circular by construction; reported for completeness only      |
| Runtime status                   | Protocol specified in code/audit_nested_cv.py; not used in the primary benchmark      |